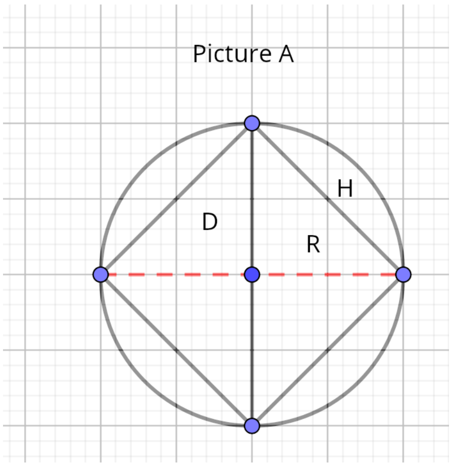

On picture A.
H = s . Its a circle chord
D = d. Red line on picture A
R = radius

new formula for hipotenuse

Math formula for diagonal of square

$d = s * \sqrt{2}$ (1)

d = diagonal of square
s = side of square

H is hipotenuse (and chord)

Using math formula, (1) above, dividing all sides by $\sqrt{2}$.

We get

$H = \frac{\mathrm{D}}{\sqrt{2}}$

D, the diameter , is 2*radius so

$D = 2 * r$

Final formula for hipotenuse

$$\text{Hipotenuse} = \frac{2 * r}{\sqrt{2}}$$

r is the base of triangle

Other considerations

new formula for radius

side of square is H (check above picture).

using pythagoras

$$\text{radius}^2 + \text{radius}^2 = \text{side of square}^2$$

$$2 * \text{radius}^2 = \text{side}^2$$

$$\text{radius} = \sqrt{\frac{side^2}{2}}$$

$$\text{radius} = \frac{\text{side}}{\sqrt{2}}$$

$$\text{radius} = \frac{H}{\sqrt{2}}$$

other formula for radius

$$\text{radius}^2 + \text{radius}^2 = 2 * \text{radius} * \text{radius}$$

$$2 * \text{radius} * \text{radius} = \text{side}^2$$

$$\text{diameter} * \text{radius} = \text{side}^2$$

$$\text{radius} = \frac{\text{side}^2}{\text{diameter}}$$

new formula for area of circle

considering new radius formula (r was calculated before).

$$r = \frac{\text{Hipotenuse}}{\sqrt{2}}$$

square r

$$r^2 = \frac{\text{Hipotenuse}^2}{2}$$

$$\text{area circle} = \pi * r^2 = \pi * \frac{\text{Hipotenuse}^2}{2}$$

new formula for perimeter of circle

$$\text{perimeter of circle} = 2\pi r = 2 * \pi * \frac{\sqrt{side^2}}{\sqrt{2}}$$

develop fraction on right side.

$$\text{perimeter of circle } 2 * \pi * \frac{\text{Hipotenuse}}{\sqrt{2}}$$